ROCK DETECTIVE

# Field Journal

Scratch it. Streak it. Sort it. Name it.



SPECIMEN LABEL

**Field scientist:** _____

**Journal started:** _____ **Specimens logged:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

# How a rock detective works

Scientists who study rocks and minerals are called **geologists** (jee-OL-oh-jists). Every rock has a story. This journal is your field notebook, and it lives in your SciOly binder.

**1**
**FIND**
Look in creek beds, gravel paths, gardens, and walls. Pick up loose rocks only.

**2**
**LOOK**
Color, luster, crystals, grains, layers, bands, holes, fossils.

**3**
**TEST**
Scratch, streak, fizz, magnet. Record every result.

**4**
**NAME IT**
Use the Mineral and Rock Collector pages and the ID tips.

**5**
**LOG IT**
Fill in a Specimen page and add it to your Collection Catalog.

## The Field Scientist Code

- [ ] Only take loose rocks from your own yard or with permission. Never take rocks from parks.
- [ ] Hammers and glass tests only with an adult, and wear safety glasses.
- [ ] Use vinegar for the fizz test. Never taste a rock or mineral.
- [ ] Wash your hands after handling rocks, and put rocks back where you found them if you can.
- [ ] Stay away from cliffs, fast water, and construction sites.
- [ ] Real specimens stay at home. Tournament binders use photos and drawings only.

## My field kit

- [ ] This journal and a pencil
- [ ] Hand lens
- [ ] Copper penny, steel nail, and a glass jar (with an adult)
- [ ] Streak plate (back of a white bathroom tile)
- [ ] Small cup of vinegar and a dropper
- [ ] Magnet and egg carton for your collection

## Journal map


| | |
|---|---|
| Tools and guides | p. 3-8 |
| Specimen pages | p. 9-14 |
| Hunts and missions | p. 15-19 |
| Collection Catalog | p. 20 |


## Where to look

**Creek beds**
smooth, rounded rocks tumbled by water, like quartz and sandstone

**Gravel paths and driveways**
crushed granite, basalt, and limestone

**Garden and landscaping**
river rock, pumice, slate stepping stones, marble chips

**Buildings and statues**
granite, marble, sandstone, and limestone walls

**Your kitchen**
salt (halite), a granite counter, a copper pot

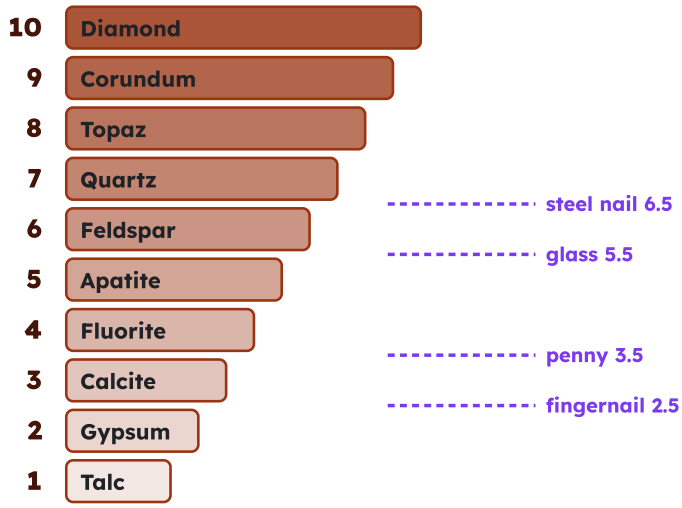
**The museum (Nov 7)**
giant crystals, gems, and meteorites up close

**Did you know?** North Carolina's state rock is granite, and NC is one of the only places on Earth where the rare green gem hiddenite is found!

# Rock or mineral?

| | |
|---|---|
| **Mineral** | 1) formed in nature, 2) not alive, 3) solid, 4) has its own chemical recipe, 5) atoms lined up in a crystal pattern. Examples: quartz, halite. |
| **Rock** | a solid mix of one or more minerals (or volcanic glass, or old plant bits). Example: granite = quartz + feldspar + mica. |
| **Igneous** | formed when melted rock cools and hardens (granite, basalt) |
| **Sedimentary** | formed when bits of rock, shells, or plants are pressed and glued together (sandstone) |
| **Metamorphic** | a rock changed by heat and pressure deep underground (slate, marble) |

## The Mohs hardness ladder




softest (1) at the bottom, hardest (10) at the top

## Scratch test

Try to scratch the specimen. If a tool scratches it, the specimen is **softer** than the tool.

- ☐ Fingernail scratches it: 2.5 or less
- ☐ Penny scratches it: about 3.5 or less
- ☐ It scratches glass: harder than 5.5
- ☐ Nail scratches it: 6.5 or less

## My Mohs memory line

Make a silly sentence with the first letters: T G C F A F Q T C D.

# Mineral tests cheat sheet

| | |
|---|---|
| **Hardness** | how hard it is to scratch. Mohs scale: 1 talc, 2 gypsum, 3 calcite, 4 fluorite, 5 apatite, 6 feldspar, 7 quartz, 8 topaz, 9 corundum, 10 diamond. |
| **Test tools** | fingernail 2.5, copper penny about 3.5, glass 5.5, steel nail or file about 6.5 |
| **Streak** | the color of the mineral's powder when rubbed on a white tile |
| **Luster** | how it shines: metallic (like metal), glassy, pearly, greasy, or dull |
| **Cleavage** | breaks along flat, smooth surfaces. Fracture: breaks unevenly or in curved shell shapes. |

## Luster: how does it shine?

metallic (like a spoon)  glassy (like a window)  pearly (like a pearl)  greasy (like butter)  dull (like chalk)

## Special clues

**Fizzes in vinegar**
calcite, limestone, marble

**Cubes that dissolve in water**
halite

**Peels in thin sheets**
mica

**Very heavy for its size**
galena, gold

**Soapy or greasy feel**
talc, graphite

**Glows under a black light**
fluorite

**Red-brown streak**
hematite

**Floats**
pumice

## ID tips

| | |
|---|---|
| **Metallic?** | Then use streak: yellow = gold, greenish black = pyrite, red-brown = hematite, gray = galena, black and greasy = graphite, copper-red = copper. |
| **Not metallic?** | Then use hardness and cleavage: sheets = mica, cubes = halite, rhombs and fizz = calcite, 8-sided = fluorite, 6-sided crystal = quartz, pink blocks = feldspar, green grains = olivine. |

**Look-alikes:** pyrite ("fool's gold") has a greenish-black streak and is hard; gold has a gold streak and is soft. Marble fizzes; quartzite does not. Slate rings when tapped; shale crumbles.

# Mineral Collector: our 16 minerals

H = hardness. Check the box when you see one for real (at home, a rock shop, or the museum). Drawings show a typical shape and color.



## Talc

*H 1 · streak white · pearly or greasy*

softest mineral; feels soapy.
*Use:* Baby powder, paint, soapstone

☐ Found: ___



## Graphite

*H 1-2 · streak black or dark gray · metallic to dull*

greasy, marks paper.
*Use:* Pencil 'lead', lubricant, batteries

☐ Found: ___



## Gypsum

*H 2 · streak white · glassy to pearly or silky*

fingernail scratches it.
*Use:* Drywall, plaster of Paris

☐ Found: ___



## Mica

*H 2-2.5 · streak white · glassy to pearly*

flexible sheets.
*Use:* Electrical insulation, makeup shimmer. Mined in NC

☐ Found: ___



## Halite

*H 2.5 · streak white · glassy*

dissolves in water.
*Use:* Table salt, road salt

☐ Found: ___



## Galena

*H 2.5 · streak lead gray · metallic*

very heavy, silver-gray cubes.
*Use:* Main ore of lead

☐ Found: ___



## Gold

*H 2.5-3 · streak golden yellow · metallic*

very heavy, bends without breaking.
*Use:* Jewelry, electronics, money

☐ Found: ___



## Copper

*H 2.5-3 · streak copper red · metallic*

copper color, may have green crust.
*Use:* Wires, pipes, coins

☐ Found: ___

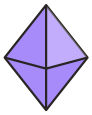


## Calcite

*H 3 · streak white · glassy*

fizzes in acid; clear pieces show double images.
*Use:* Cement, antacids

☐ Found: ___



## Fluorite

*H 4 · streak white · glassy*

purple, green, or blue cubes; may glow under black light.
*Use:* Steel making, fluoride

☐ Found: ___



## Hematite

*H 5.5-6.5 · streak red-brown · metallic or dull*

red-brown streak even when silver-black.
*Use:* Main ore of iron

☐ Found: ___



## Feldspar (pink)

*H 6 · streak white · glassy*

pink, blocky; in granite.
*Use:* Ceramics, glass

☐ Found: ___



## Pyrite

*H 6-6.5 · streak greenish black · metallic*

brassy cubes; 'fool's gold'.
*Use:* Source of sulfur

☐ Found: ___



## Olivine

*H 6.5-7 · streak white · glassy*

olive green grains, often in basalt.
*Use:* Gem called peridot

☐ Found: ___



## Quartz (crystal)

*H 7 · streak white · glassy*

6-sided crystals, scratches glass.
*Use:* Glass, electronics, sand. NC quartz is used for computer chips

☐ Found: ___



## Corundum

*H 9 · streak white · glassy to shiny (adamantine)*

very hard 6-sided barrels.
*Use:* Rubies and sapphires, sandpaper

☐ Found: ___

# Rock Collector: our 17 rocks

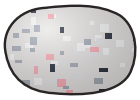
Check the box when you find one. Igneous rocks cooled from melted rock. Sedimentary rocks are bits pressed and glued together. Metamorphic rocks were changed by heat and pressure.

## Igneous



### Basalt

Dark, heavy, tiny crystals. Lava that cooled fast.
*Use:* road gravel ☐



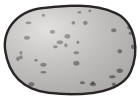
### Granite

Big crystals you can see: pink, white, black. Cooled slowly underground.
*Use:* countertops; NC's state rock ☐



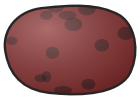
### Obsidian

Shiny black glass with curved breaks. Cooled super fast.
*Use:* arrowheads, scalpels ☐



### Pumice

Light gray, full of tiny holes. It floats!
*Use:* foot scrubbers ☐



### Scoria

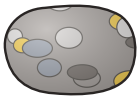
Red-black with big holes. Does not float.
*Use:* landscaping, grills ☐

## Sedimentary



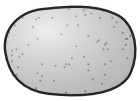
### Bituminous coal

Black, light, shiny and dull layers. Old swamp plants.
*Use:* fuel for electricity ☐



### Conglomerate

Rounded pebbles glued together.
*Use:* decorative stone ☐



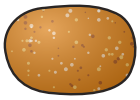
### Limestone (chemical)

Gray or white, fine. Fizzes in vinegar.
*Use:* cement ☐



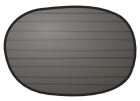
### Limestone (fossil)

Shells and fossils glued together. Fizzes.
*Use:* building stone ☐



### Sandstone

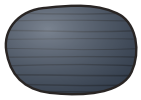
Feels like sandpaper; may show layers.
*Use:* building stone ☐



### Shale

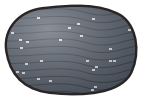
Thin, flat layers of hardened mud. Breaks easily.
*Use:* bricks ☐

## Metamorphic



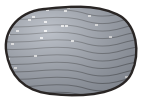
### Slate

Flat, hard sheets that ring when tapped. Parent: shale.
*Use:* roofs, chalkboards ☐



### Phyllite

Satin shine, wavy layers. Parent: slate.
*Use:* decorative stone ☐



### Schist

Sparkly mica flakes in wavy layers. Parent: phyllite.
*Use:* garden stone ☐



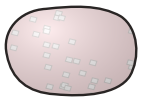
### Gneiss

Light and dark stripes (bands). Parent: granite or schist.
*Use:* countertops ☐



### Marble

Sugary crystals, fizzes. Parent: limestone.
*Use:* statues ☐

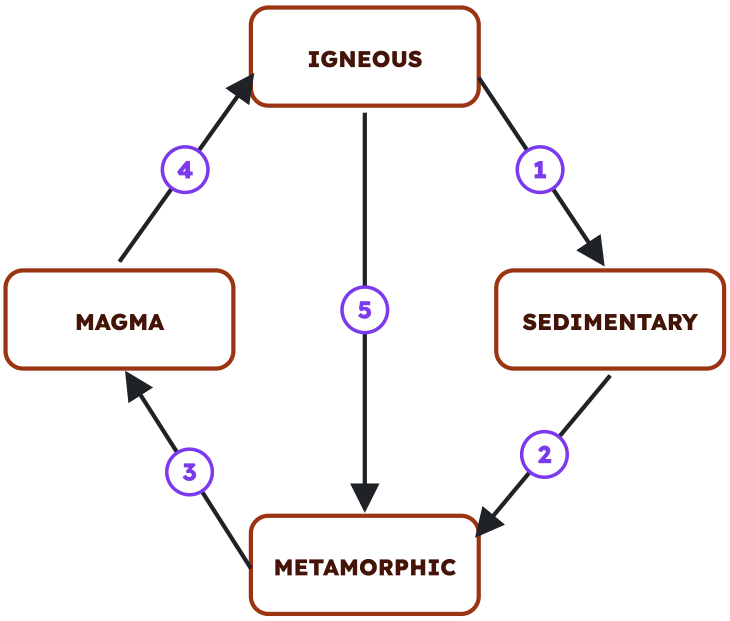


### Quartzite

Very hard, sugary, no fizz. Parent: sandstone.
*Use:* countertops ☐

# The rock cycle

Rocks change from one kind to another over millions of years. Write what happens on each numbered arrow. Some arrows need more than one word.




1. _____
2. _____
3. _____
4. _____
5. _____

**cooling** **melting** **weathering** **erosion** **deposition** **compaction** **cementation** **heat and pressure**

Arrow 5 goes straight from igneous to metamorphic. Why can any rock skip steps?

## My rock's journey

Pick a rock from your collection. Write its story as it travels the cycle, like "I was a grain of sand on a beach…"

# Scratch and Streak Lab

Test the specimens in your collection or your team's test kit. Use this page with Independent Lab I1 (Kitchen Mineral Lab). Write Y if the tool scratches the specimen and N if it does not.

| Specimen | Nail 2.5 | Penny 3.5 | Scratches glass? | Steel 6.5 | Hardness about | Streak color | Fizz? |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| | | | | | | | |
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| | | | | | | | |
| | | | | | | | |

**Tape or draw your streaks here**

## What I found out

Which specimen was hardest? How do you know?

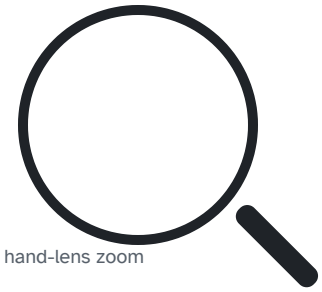
# Specimen #1

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

**Date** _____ **Time** _____ **Place** _____

**Found:** ☐ backyard ☐ creek bed ☐ gravel or path ☐ garden ☐ rock shop or kit ☐ museum ☐ other

**Sketch it BIG. Show crystals, grains, layers, or holes.**



hand-lens zoom

Real size: about _____ cm

## TESTS

Color _____

Luster _____

Streak _____

Hardness _____

Cleavage? _____

## CLUES I SEE

- ☐ crystals
- ☐ grains
- ☐ layers
- ☐ bands
- ☐ holes
- ☐ fossils
- ☐ shiny flakes

## IT IS A

- ☐ mineral
- ☐ igneous rock
- ☐ sedimentary rock
- ☐ metamorphic rock

**ROCK OR MINERAL?** _____

**MY ID** _____

**A USE** _____

**FIZZ OR MAGNET?** _____

**MY BEST CLUE** _____

**How did it form?**

**I wonder...**

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

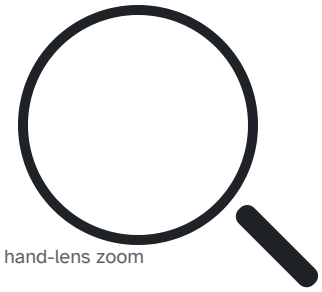
# Specimen #2

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

**Date** _____ **Time** _____ **Place** _____

**Found:** ☐ backyard ☐ creek bed ☐ gravel or path ☐ garden ☐ rock shop or kit ☐ museum ☐ other

**Sketch it BIG. Show crystals, grains, layers, or holes.**



hand-lens zoom

Real size: about _____ cm

## TESTS

Color _____

Luster _____

Streak _____

Hardness _____

Cleavage? _____

## CLUES I SEE

- ☐ crystals
- ☐ grains
- ☐ layers
- ☐ bands
- ☐ holes
- ☐ fossils
- ☐ shiny flakes

## IT IS A

- ☐ mineral
- ☐ igneous rock
- ☐ sedimentary rock
- ☐ metamorphic rock

**ROCK OR MINERAL?** _____

**MY ID** _____

**A USE** _____

**FIZZ OR MAGNET?** _____

**MY BEST CLUE** _____

**How did it form?**

**I wonder...**

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



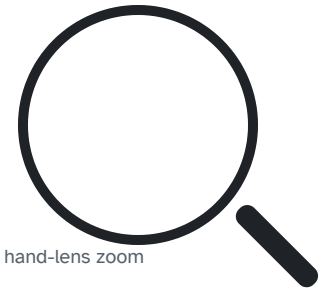
# Specimen #3

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

**Date** _____ **Time** _____ **Place** _____

**Found:** ☐ backyard ☐ creek bed ☐ gravel or path ☐ garden ☐ rock shop or kit ☐ museum ☐ other

**Sketch it BIG. Show crystals, grains, layers, or holes.**



hand-lens zoom

Real size: about _____ cm

## TESTS

Color _____

Luster _____

Streak _____

Hardness _____

Cleavage? _____

## CLUES I SEE

- ☐ crystals
- ☐ grains
- ☐ layers
- ☐ bands
- ☐ holes
- ☐ fossils
- ☐ shiny flakes

## IT IS A

- ☐ mineral
- ☐ igneous rock
- ☐ sedimentary rock
- ☐ metamorphic rock

**ROCK OR MINERAL?** _____

**MY ID** _____

**A USE** _____

**FIZZ OR MAGNET?** _____

**MY BEST CLUE** _____

**How did it form?**

**I wonder...**

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



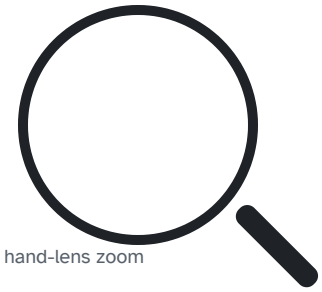
# Specimen #4

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

**Date** _____ **Time** _____ **Place** _____

**Found:** ☐ backyard ☐ creek bed ☐ gravel or path ☐ garden ☐ rock shop or kit ☐ museum ☐ other

**Sketch it BIG. Show crystals, grains, layers, or holes.**



hand-lens zoom

Real size: about _____ cm

## TESTS

Color _____

Luster _____

Streak _____

Hardness _____

Cleavage? _____

## CLUES I SEE

- ☐ crystals
- ☐ grains
- ☐ layers
- ☐ bands
- ☐ holes
- ☐ fossils
- ☐ shiny flakes

## IT IS A

- ☐ mineral
- ☐ igneous rock
- ☐ sedimentary rock
- ☐ metamorphic rock

**ROCK OR MINERAL?** _____

**MY ID** _____

**A USE** _____

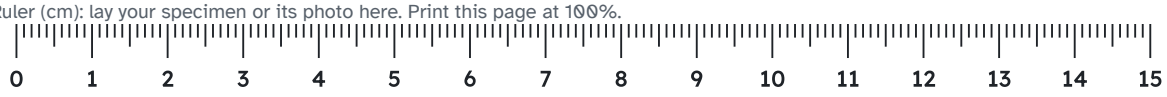
**FIZZ OR MAGNET?** _____

**MY BEST CLUE** _____

**How did it form?**

**I wonder...**

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



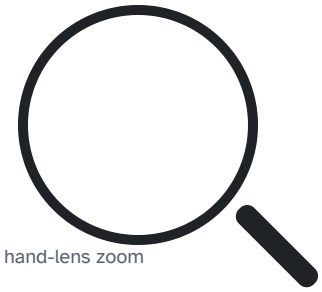
# Specimen #5

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

**Date** _____ **Time** _____ **Place** _____

**Found:** ☐ backyard ☐ creek bed ☐ gravel or path ☐ garden ☐ rock shop or kit ☐ museum ☐ other

**Sketch it BIG. Show crystals, grains, layers, or holes.**



hand-lens zoom

Real size: about _____ cm

## TESTS

Color _____

Luster _____

Streak _____

Hardness _____

Cleavage? _____

## CLUES I SEE

- ☐ crystals
- ☐ grains
- ☐ layers
- ☐ bands
- ☐ holes
- ☐ fossils
- ☐ shiny flakes

## IT IS A

- ☐ mineral
- ☐ igneous rock
- ☐ sedimentary rock
- ☐ metamorphic rock

**ROCK OR MINERAL?** _____

**MY ID** _____

**A USE** _____

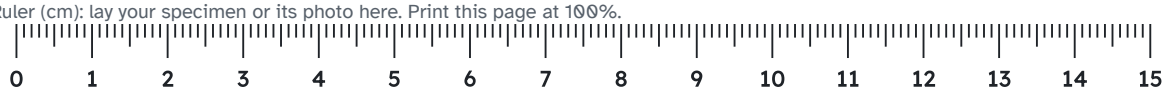
**FIZZ OR MAGNET?** _____

**MY BEST CLUE** _____

**How did it form?**

**I wonder...**

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



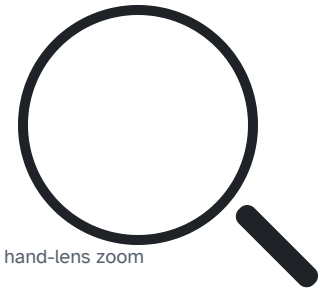
# Specimen #6

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

**Date** _____ **Time** _____ **Place** _____

**Found:** ☐ backyard ☐ creek bed ☐ gravel or path ☐ garden ☐ rock shop or kit ☐ museum ☐ other

**Sketch it BIG. Show crystals, grains, layers, or holes.**



hand-lens zoom

Real size: about _____ cm

## TESTS

Color _____

Luster _____

Streak _____

Hardness _____

Cleavage? _____

## CLUES I SEE

- ☐ crystals
- ☐ grains
- ☐ layers
- ☐ bands
- ☐ holes
- ☐ fossils
- ☐ shiny flakes

## IT IS A

- ☐ mineral
- ☐ igneous rock
- ☐ sedimentary rock
- ☐ metamorphic rock

**ROCK OR MINERAL?** _____

**MY ID** _____

**A USE** _____

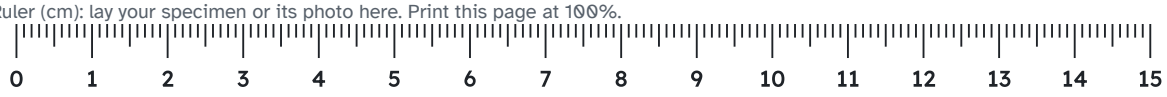
**FIZZ OR MAGNET?** _____

**MY BEST CLUE** _____

**How did it form?**

**I wonder...**

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



# Minerals at Home

**Mission:** with an adult, find at least 5 things in your home that come from rocks or minerals. Two are done for you.

| Thing I found | Rock or mineral in it | Room | What it does |
|---|---|---|---|
| *Salt shaker* | *halite* | *kitchen* | *flavor food* |
| *Pencil* | *graphite* | *desk* | *writing* |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## Idea list

baby powder (talc), drywall (gypsum), toothpaste (fluorite), wires (copper), phone (gold, copper, quartz), windows (quartz sand), antacid tablets (calcite), a granite or marble counter, jewelry, bricks (shale).

## Why do minerals matter to money and jobs?

# Rock Walk

**Mission:** take a walk around your neighborhood, school, or downtown with an adult. Look at walls, steps, statues, gravel, and garden rocks. What rocks did people build with?

| Where | What it is made of | Rock family | Clue I used |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

**Sketch the coolest rock I saw**

## Look for

- **Granite** steps, curbs, and monuments (speckled)
- **Marble** statues and old buildings (white, sugary)
- **Slate** roofs and stepping stones (flat sheets)
- **Brick** is baked shale and clay
- **Concrete** is made with limestone cement

**Why would a builder pick that rock?**

# Museum Day: Rocks, Gems, and Crystals

Saturday, November 7. Meet at 10 am at the Daily Planet Cafe. Look for minerals from our list, giant crystals, and North Carolina gems.

## Minerals from our list I saw

| Mineral | A clue I noticed |
|---|---|
| | |
| | |
| | |
| | |
| | |

## A question for a museum scientist

## Sketch the biggest crystal I saw

Name: _____ Color: _____

## Best thing I learned today

# Field Missions

One mission for each Geology Rocks! meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge.

| When | Mission | Badge | Done | Initials |
|---|---|---|---|---|
| **After Fri Oct 2**<br>Lesson 1 | Find 5 things at home made from minerals (p. 15). | Mineral Spotter | ☐ | |
| **After Fri Nov 6**<br>Lesson 2 | Kitchen Mineral Lab (Lab I1) and 3 rows of the Scratch and Streak Lab. | Scratch Tester | ☐ | |
| **Sat Nov 7**<br>Museum | Fill in the Museum Day page. | Museum Explorer | ☐ | |
| **After Fri Dec 11**<br>Lesson 3 | Rock Cycle Journey (Lab I2) and label the rock cycle (p. 7). | Rock Cycle Rider | ☐ | |
| **After Fri Jan 29**<br>Lesson 4 | Collect 3 rocks, test them, and sort them into families on Specimen pages. | Rock Hound | ☐ | |
| **Fri Feb 26**<br>Lesson 5 | Cover the names on p. 5-6 and name 10 specimens in under 2 minutes. | Speed Spotter | ☐ | |
| **Anytime**<br>Bonus | Do a streak test on a real specimen. Take a Rock Walk (p. 16). | Streak Seeker / Rock Walker | ☐ | |

## Tournament day: Sat Mar 13, Chatham Regional

- ☐ 2 hand lenses
- ☐ Your 1-inch binder (photos, no real samples)
- ☐ 2 sharp pencils

- **Metallic?** Think streak. **Not metallic?** Think hardness and cleavage.
- Look for crystals, grains, layers, bands, holes, and fossils.
- Answer both parts of every station before time is called.

# My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Rock Detective**! Your Team Badge Passport has the same badges.



**Mineral Spotter**

5 minerals found at home

Initials: _____



**Scratch Tester**

Kitchen Mineral Lab + scratch tests

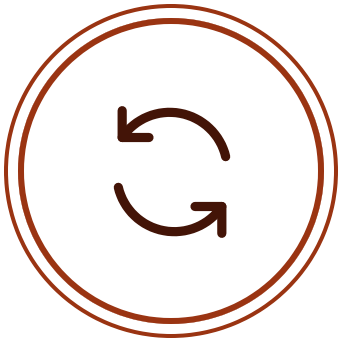
Initials: _____



**Museum Explorer**

Finish the Museum Day page

Initials: _____



**Rock Cycle Rider**

Label the rock cycle + Lab I2

Initials: _____



**Rock Hound**

Test and sort 3 real rocks

Initials: _____



**Speed Spotter**

10 specimens in under 2 minutes

Initials: _____



**Streak Seeker**

Streak test a real specimen

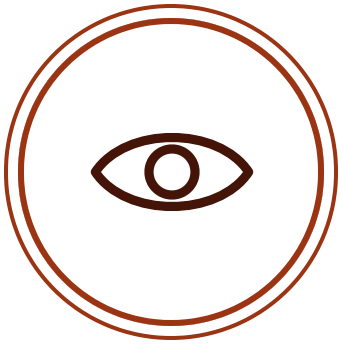
Initials: _____



**Rock Walker**

Take a Rock Walk

Initials: _____



**Field Scientist**

Log 4 Specimen pages

Initials: _____

# My Collection Catalog

Number each specimen in your egg carton or box and write it here. Photos and drawings count too.

| # | Date | Name | Rock or mineral | Family | Where found |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |
| 11 | | | | | |
| 12 | | | | | |
| 13 | | | | | |
| 14 | | | | | |
| 15 | | | | | |
| 16 | | | | | |
| 17 | | | | | |
| 18 | | | | | |
| 19 | | | | | |
| 20 | | | | | |
| 21 | | | | | |
| 22 | | | | | |
| 23 | | | | | |
| 24 | | | | | |